

Lecture 2: Introduction to Convexity

Instructor Notes & Proofs

ISE 5406: Nonlinear Programming

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Contents

1 Linear Algebra Review

1.1 Blank Slide After Halfspaces (Slides 10–11)

Board Work: Visualizing Halfspaces

Suggested content:

- Draw $\mathbb{R}^2$ example with a hyperplane $a_1x_1 + a_2x_2 = b$
- Shade the two halfspaces on either side
- Show how $\mathbf{a}$ is the normal vector pointing toward $\mathbf{a}^T \mathbf{x} \geq b$
- Example: $x_1 + 2x_2 = 4$
 - Halfspace $x_1 + 2x_2 \leq 4$ (below/left of line)
 - Halfspace $x_1 + 2x_2 \geq 4$ (above/right of line)

1.2 Types of Combinations — Visual Summary

Board Work: Comparison of Combinations

Draw a table or diagram comparing:

Type	Coefficients λ_j	Geometric Object
Linear	$\lambda_j \in \mathbb{R}$	Entire subspace
Conic	$\lambda_j \geq 0$	Cone from origin
Affine	$\sum \lambda_j = 1$	Affine subspace (translated)
Convex	$\lambda_j \geq 0, \sum \lambda_j = 1$	Convex hull

Visual example with 2 points $\mathbf{a}_1, \mathbf{a}_2$:

- Linear: entire line through both (extended infinitely)
- Affine: same as linear for 2 points
- Conic: cone with vertex at origin, rays through $\mathbf{a}_1$ and $\mathbf{a}_2$
- Convex: line segment $[\mathbf{a}_1, \mathbf{a}_2]$

2 Convex Sets

2.1 Blank Slides After Convex Set Examples (Slides 17–18)

Board Work: Verifying Convexity of Sets

Work through the “Which sets are convex?” examples:

1. $\mathbb{R}^n$ and $\emptyset$: Both convex (vacuously true for $\emptyset$).

2. **Unit disk** $\{\mathbf{x} \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$:

- Take $\mathbf{x}_1, \mathbf{x}_2$ in the disk, so $\|\mathbf{x}_1\|_2 \leq 1$ and $\|\mathbf{x}_2\|_2 \leq 1$
- For $\lambda \in [0, 1]$: $\|\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2\|_2 \leq \lambda \|\mathbf{x}_1\|_2 + (1 - \lambda) \|\mathbf{x}_2\|_2 \leq 1$
- Open disk: same argument, convex

3. $\{\mathbf{x} : \mathbf{Ax} = \mathbf{b}\}$:

- If $\mathbf{Ax}_1 = \mathbf{b}$ and $\mathbf{Ax}_2 = \mathbf{b}$, then $\mathbf{A}(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) = \lambda \mathbf{b} + (1 - \lambda) \mathbf{b} = \mathbf{b}$ ✓

4. $\{\mathbf{x} : \mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq 0\}$: Intersection of convex sets $\rightarrow$ convex.

5. **Hyperplanes and halfspaces**: Special cases of linear constraints.

6. $\{\mathbf{x} \in \mathbb{R}_+^2 : x_1 x_2 = 0\}$:

- This is the union of the two axes (not including interior)
- Take $(1, 0)$ and $(0, 1)$: midpoint $(\frac{1}{2}, \frac{1}{2})$ has $x_1 x_2 = \frac{1}{4} \neq 0$
- **NOT convex!**

7. $\{\mathbf{x} \in \mathbb{R}^2 : x_1^2 - x_1 x_2^2 + x_2^4 + 1 \geq 0\}$:

- This is actually convex! (sublevel set of a convex function — verify by Hessian if time permits)
- Or note: $x_1^2 - x_1 x_2^2 + x_2^4 + 1 = (x_1 - \frac{x_2^2}{2})^2 + \frac{3x_2^4}{4} + 1 \geq 1 > 0$ always
- So this is all of $\mathbb{R}^2$!

3 Convex and Concave Functions

3.1 Blank Slides After Function Definitions (Slides 28–29)

Board Work: Visualizing Convex/Concave Functions

Draw the “chord above graph” picture:

- For convex f : the line segment (chord) connecting $(x_1, f(x_1))$ to $(x_2, f(x_2))$ lies *above* the graph
- The inequality $f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$ says exactly this

Examples to sketch:

- $f(x) = x^2$ (convex parabola)
- $f(x) = e^x$ (convex exponential)
- $f(x) = \ln(x)$ (concave logarithm)
- $f(x) = -x^2$ (concave parabola)

Why affine functions are both convex and concave:

$$f(\lambda x_1 + (1 - \lambda)x_2) = a(\lambda x_1 + (1 - \lambda)x_2) + b = \lambda(ax_1 + b) + (1 - \lambda)(ax_2 + b)$$

Equality holds! So both $\leq$ and $\geq$ are satisfied.

3.2 Proof: Level Sets of Convex Functions are Convex (Slide 32)

Proof to Present

Lemma: Let S be a convex set and $f : S \rightarrow \mathbb{R}$ be a convex function. For any $\alpha \in \mathbb{R}$, the level set $S_\alpha = \{\mathbf{x} \in S : f(\mathbf{x}) \leq \alpha\}$ is convex.

Proof:

1. Take any $\mathbf{x}_1, \mathbf{x}_2 \in S_\alpha$ and $\lambda \in [0, 1]$.
2. We need to show $\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 \in S_\alpha$.
3. Since S is convex, $\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 \in S$. ✓
4. Since f is convex:

$$f(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2) \leq \lambda f(\mathbf{x}_1) + (1 - \lambda)f(\mathbf{x}_2)$$

5. Since $\mathbf{x}_1, \mathbf{x}_2 \in S_\alpha$, we have $f(\mathbf{x}_1) \leq \alpha$ and $f(\mathbf{x}_2) \leq \alpha$.
6. Therefore:

$$f(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2) \leq \lambda\alpha + (1 - \lambda)\alpha = \alpha$$

7. So $\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 \in S_\alpha$. □

Remark. The converse is **false**! A function can have convex level sets without being convex.

Counterexample: $f(x) = x^3$ on $\mathbb{R}$. Level sets $\{x : x^3 \leq \alpha\} = (-\infty, \alpha^{1/3}]$ are convex (intervals), but $f(x) = x^3$ is neither convex nor concave.

Such functions are called **quasiconvex**.

4 Convex Hulls

4.1 Carathéodory's Theorem (Slide 37)

The proof is assigned as reading (Theorem 2.1.6 of Bazaraa et al.), but here's an outline:

Proof Outline: Carathéodory's Theorem

Theorem: Let $S \subseteq \mathbb{R}^n$. If $\mathbf{x} \in \text{conv}(S)$, then $\mathbf{x}$ can be written as a convex combination of at most $n + 1$ points from S .

Proof idea:

1. By definition, $\mathbf{x} = \sum_{j=1}^k \lambda_j \mathbf{x}_j$ with $\lambda_j > 0$, $\sum \lambda_j = 1$, and $\mathbf{x}_j \in S$.
2. If $k \leq n + 1$, we're done.
3. If $k > n + 1$, we show we can reduce the number of points by one while keeping $\mathbf{x}$ in the convex hull.
4. **Key step:** The vectors $\mathbf{x}_2 - \mathbf{x}_1, \mathbf{x}_3 - \mathbf{x}_1, \dots, \mathbf{x}_k - \mathbf{x}_1$ are $k - 1 > n$ vectors in $\mathbb{R}^n$, so they must be linearly dependent.
5. This means $\exists$ coefficients μ_j (not all zero) with $\sum_{j=1}^k \mu_j = 0$ and $\sum_{j=1}^k \mu_j \mathbf{x}_j = \mathbf{0}$.
6. For small enough $t > 0$: $\mathbf{x} = \sum_{j=1}^k (\lambda_j - t\mu_j) \mathbf{x}_j$ still has $\sum (\lambda_j - t\mu_j) = 1$.
7. Choose t so that one coefficient becomes zero $\rightarrow$ representation with $k - 1$ points.
8. Repeat until $k \leq n + 1$. □

5 Introduction to Convex Analysis

5.1 Extreme Points and Rays

Board Work: Extreme Points

Examples to draw:

- Square in $\mathbb{R}^2$: 4 extreme points (corners)
- Disk in $\mathbb{R}^2$: every boundary point is extreme!
- Simplex in $\mathbb{R}^n$: vertices are extreme points
- Halfspace: no extreme points (unbounded)

Is a ray convex?

- Ray: $\{\mathbf{x}_0 + \lambda \mathbf{d} : \lambda \geq 0\}$
- Take two points: $\mathbf{x}_0 + \lambda_1 \mathbf{d}$ and $\mathbf{x}_0 + \lambda_2 \mathbf{d}$ with $\lambda_1, \lambda_2 \geq 0$
- Convex combination: $\mu(\mathbf{x}_0 + \lambda_1 \mathbf{d}) + (1 - \mu)(\mathbf{x}_0 + \lambda_2 \mathbf{d}) = \mathbf{x}_0 + (\mu\lambda_1 + (1 - \mu)\lambda_2)\mathbf{d}$
- Since $\mu\lambda_1 + (1 - \mu)\lambda_2 \geq 0$, this is in the ray. ✓
- **Yes, rays are convex!**

5.2 Recession Directions

Board Work: Recession Cone Example

For the set $X = \{\mathbf{x} \in \mathbb{R}_+^2 : x_1 - x_2 \geq -2, x_1 + x_2 \geq 1\}$:

Draw the feasible region (intersection of halfspaces).

Recession directions: A direction $\mathbf{d}$ is a recession direction if for every $\mathbf{x}_0 \in X$ and every $\lambda \geq 0$, we have $\mathbf{x}_0 + \lambda \mathbf{d} \in X$.

For a polyhedron $X = \{\mathbf{x} : \mathbf{Ax} \leq \mathbf{b}\}$:

$$\text{Recession cone} = \{\mathbf{d} : \mathbf{Ad} \leq \mathbf{0}\}$$

For this example:

- $x_1 \geq 0 \Rightarrow d_1 \geq 0$
- $x_2 \geq 0 \Rightarrow d_2 \geq 0$
- $x_1 - x_2 \geq -2 \Rightarrow d_1 - d_2 \geq 0$
- $x_1 + x_2 \geq 1 \Rightarrow d_1 + d_2 \geq 0$ (automatic if $d_1, d_2 \geq 0$)

Recession cone: $\{\mathbf{d} \in \mathbb{R}^2 : d_1 \geq d_2 \geq 0\}$

6 Operations Preserving Convexity — Proofs (Optional)

These proofs are straightforward and can be presented if time permits or assigned as exercises.

Proof: Intersection Preserves Convexity

Let S_1, S_2 be convex sets. We show $S_1 \cap S_2$ is convex.

Take $\mathbf{x}_1, \mathbf{x}_2 \in S_1 \cap S_2$ and $\lambda \in [0, 1]$.

- Since $\mathbf{x}_1, \mathbf{x}_2 \in S_1$ and S_1 is convex: $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S_1$.
- Since $\mathbf{x}_1, \mathbf{x}_2 \in S_2$ and S_2 is convex: $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S_2$.
- Therefore $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S_1 \cap S_2$. □

Proof: Affine Image Preserves Convexity

Let S be convex and consider $T = \{\mathbf{Ax} + \mathbf{b} : \mathbf{x} \in S\}$ for $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$.

Take $\mathbf{y}_1, \mathbf{y}_2 \in T$. Then $\mathbf{y}_1 = \mathbf{Ax}_1 + \mathbf{b}$ and $\mathbf{y}_2 = \mathbf{Ax}_2 + \mathbf{b}$ for some $\mathbf{x}_1, \mathbf{x}_2 \in S$.

For $\lambda \in [0, 1]$:

$$\begin{aligned} \lambda \mathbf{y}_1 + (1 - \lambda) \mathbf{y}_2 &= \lambda(\mathbf{Ax}_1 + \mathbf{b}) + (1 - \lambda)(\mathbf{Ax}_2 + \mathbf{b}) \\ &= \mathbf{A}(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) + \mathbf{b} \end{aligned}$$

Since S is convex, $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S$, so $\lambda \mathbf{y}_1 + (1 - \lambda) \mathbf{y}_2 \in T$. □

7 Looking Ahead: Preview of Next Topics

Teaser for Next Lecture

Topics to preview at end of lecture (if time):

- **Separation theorems:** Can we always separate a point from a convex set with a hyperplane?
- **Supporting hyperplanes:** Hyperplanes that “touch” a convex set at a boundary point
- **Characterizations of convex functions:** First-order and second-order conditions
- **Why convexity matters for optimization:** Local optimum = global optimum!